

Aaron Chu

669.946.5814 | aaronchu022@gmail.com | linkedin.com/in/achu02 | github.com/aarocho

EXPERIENCE

Software Engineering Intern

June 2026 – Aug. 2026

Magnefy

Palo Alto, CA

- Achieved 0.991 macro F1 with **XGBoost** classifying 3 partial-discharge types on 27K lab pulses (grouped 5-fold cross-validation); benchmarked against logistic regression and MLP
- Replaced an **ML** anomaly gate with a rule-based arcing/digital-noise screen on raw samples, reaching 100% arcing recall (695/695) at 1.48% false positives in production
- Produced per-asset health summaries by building asset-monitoring, data-**drift**, and **trend**-history modules and a Dynamic Health Index encoder
- Built an automated **ML** inference pipeline on top of existing **S3** IoT storage (**S3**→**SQS**, 4 **Railway** services) that processed 598 captures (193K waveforms) from 9 units

Lead Systems Integration Engineer

Sept. 2025 – Present

University Nanosatellite Program

New Brunswick, NJ

- Increased data reliability by building distributed **Python** services on **Linux/Raspberry Pi** (**multithreaded** polling, structured sensor ingestion) as lead of a 10-member systems integration team
- Improved reliability of multi-sensor data collection on embedded systems by building integration tests and diagnostic tooling for hardware interfaces and analyzing failure cases
- Authored 3 Critical Design Review verification test procedures (communications, attitude determination and control, thermal) for SPICESat, a 6-unit **CubeSat**, tracing 20+ requirements to verification methods

Drone Support Intern

June 2025 – Aug. 2025

Integem

Cupertino, CA

- Implemented 3 PID loops (altitude, forward, yaw) on an **ESP32-S3** balloon drone using time-of-flight, optical-flow, and IMU sensors, with anti-windup and output clamping; tuned gains from logged flight data
- Split autonomous flight between a **MaixCAM** vision processor and an **ESP32** flight controller over a JSON/**UART** link, fusing **AprilTag** headings with IMU yaw to cancel drift
- Built high-altitude balloon telemetry: **Arduino C++** firmware sent GPS/pressure data over **LoRaWAN/Helium** with geofenced switching across 5 radio regions, decoded into **InfluxDB/Grafana** for live tracking

PROJECTS

DeskPartner

github.com/aarocho/DeskPartner

- Architected the 5-module stack (arm control, vision calibration, **VLM** planner, data collection, training) for a 5-person team; earned Honorable Mention at the Embodied Metal Hackathon
- Ported **Rerun**'s robotics logging toolkit (record, catalog, query, replay) to an unsupported 7-joint arm, open-sourced it, and led the team's Rerun bounty integration
- Built arm motion primitives and **ArUco** calibration integrated with a **VLM** perception-to-action loop (dry-run only)
- Wrote data-collection tooling designed for the team's vision-language-action policy (**MolmoAct2**): a validator for extra channels and missing cameras, a camera-drift check, and Rerun-to-**LeRobot** export

PocketLab

github.com/ja-zoe/pocket-lab

- Built the **React/TypeScript** frontend (dashboard, history, login) with real-time **Recharts/Three.js** visualizations, wired to **ESP32-S3** sensors through a **FastAPI WebSocket** backend (team of 4)
- Wrote a rolling mean and standard-deviation spike filter (adjustable window and threshold) that interpolates over outliers on 9 sensor channels, with **React** hooks and controls
- Built AI experiment summaries: an **Express** endpoint computes per-sensor statistics, prompts **Gemini 2.0 Flash** for commentary, and falls back to rule-based text if the API key is missing or the call fails

TECHNICAL SKILLS

Languages: Python, C++, C, Java, TypeScript/JavaScript, SQL

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, NumPy

Backend, Web, and Cloud: AWS, FastAPI, Node.js, PostgreSQL, Supabase, React, Expo, Docker

Tools: Git, Linux, pytest, Streamlit, Pydantic

EDUCATION

Rutgers University

New Brunswick, NJ

BS Computer Engineering

May 2028

- **Coursework:** Software Engineering, Data Structures, **Distributed Deep Learning**, Computer Architecture, Discrete Mathematics, Probability & Random Processes